Auteur: Karanjot Singh
Studierichting: Informatica
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$$\int \frac{x}{\sqrt{x^2+5}}dx$$

Stel $u = x^2 + 5$

$\Rightarrow du = 2xdx$

$\Rightarrow \dfrac{1}{2}du = xdx$

$= \dfrac{1}{2}\displaystyle\int \frac{1}{\sqrt{u}}du$

$= \dfrac{1}{2}\displaystyle\int u^{-\frac{1}{2}}du$

$= \dfrac{1}{2} \cdot \dfrac{u^{\frac{1}{2}}}{\frac{1}{2}} + C$

$= u^{\frac{1}{2}} + C$

$= \sqrt{x^2+5} + C$

$\Rightarrow \displaystyle\int \frac{x}{\sqrt{x^2+5}}dx = \sqrt{x^2+5} + C$

References